

Two-Sample Tests for Variance

CSU, Chico Math 351

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F Distribution Reminder

Let $\chi_{n_1}^2$ and $\chi_{n_2}^2$ be two independent chi square random variables with n_1 and n_2 degrees of freedom, respectively. A random variable of the form $\frac{\chi_{n_1}^2}{\chi_{n_2}^2}$ is said to have an **F distribution** with n_1 and n_2 degrees of freedom.

This is a right-skewed distribution, much like the χ^2 Distribution, and can be used to compare variances.

Applying F-test to Variance

Consider two populations that are expected to have similar means, but may have very different variances.

Let $X_1, \dots, X_n$ represent **n** values from a Normal distribution with mean μ_X and variance σ_X^2 , and $Y_1, \dots, Y_m$ represent **m** values from a Normal distribution with mean μ_Y and variance σ_Y^2 . We can perform a hypothesis test to compare the variances of these two populations:

$$H_0 : \sigma_X^2 = \sigma_Y^2$$

$$H_1 : \sigma_X^2 \neq \sigma_Y^2$$

Using the sample variances s_X^2 and s_Y^2 , we can define the test statistic $F = \frac{s_Y^2}{s_X^2}$.

NOTE: The degrees of freedom order matters! The first value will be for the **numerator**, and the second value will be for the **denominator**. In the above case, it would be **m-1** (for sample Y), then **n-1** (for sample X).

p-values

We can use the `pf()` function to calculate p-values for the F Distribution, just as we used `pnorm()` and `pt()` for their respective distributions.

As with the χ^2 distribution, this is **NOT** a symmetric distribution, and $\alpha/2$ and $1 - \alpha/2$ percentiles will not be opposites.

1. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 < \sigma_Y^2$, use:

$$\text{p-value} = \text{pf}(F, m-1, n-1)$$

2. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 > \sigma_Y^2$, use:

$$\text{p-value} = 1 - \text{pf}(F, m-1, n-1)$$

3. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 \neq \sigma_Y^2$, use:

$$\text{p-value} = 2 * \min(\text{pf}(F, m-1, n-1), 1 - \text{pf}(F, m-1, n-1))$$

Alternative Approach: When to Reject

Alternatively, if we just care about whether we reject H_0 or not, we can use the critical values from earlier to define our cutoffs:

1. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 < \sigma_Y^2$:

Reject H_0 if $F \leq F_{\alpha, m-1, n-1}^*$.

2. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 > \sigma_Y^2$:

Reject H_0 if $F \geq F_{1-\alpha, m-1, n-1}^*$.

3. To test $H_0 : \sigma_X^2 = \sigma_Y^2$ vs. $H_1 : \sigma_X^2 \neq \sigma_Y^2$:

Reject H_0 if $F \leq F_{\alpha/2, m-1, n-1}^*$ OR $F \geq F_{1-\alpha/2, m-1, n-1}^*$

where $F_{\alpha/2, m-1, n-1}^* = \text{qf}(\alpha/2, m-1, n-1)$.

Example (from *Larsen & Marx*, p.472)

Electroencephalograms monitor changes in electrical activity in the brain, such as alpha waves. An experiment to test sensory deprivation over an extended period of time tested two groups of 10 inmates in a Canadian prison—one placed in solitary confinement, and another left in their own cells as a control. Alpha-wave frequencies were recorded for all individuals after seven days.

Using the provided data set, conduct a hypothesis test with the F distribution to determine whether there was a statistically significant difference in alpha wave frequency variability for the two groups.

Data: [alphawave.csv](#)

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References

references I

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Adapted from notes by Dr. Roualdes.